

SEVERITY OF THE SIGN PROBLEM IN ZERO-DIMENSIONAL SUPERSYMMETRIC QFTs

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Abstract

Lattice field theory is one of the most prominent non-perturbative approaches to solving quantum field theories with strong interactions. The complex Langevin method, based on stochastic quantization, is a promising tool for the simulations of lattice regularized quantum field theories with complex actions. In such a scenario, the standard Monte Carlo methods encounter a complex phase problem or a sign problem and are rendered inapplicable. This report studies the severity of the sign problem in zero-dimensional supersymmetric quantum field theories. We also briefly discuss supersymmetric quantum mechanics.

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1 Introduction

Field theories are systems with infinitely many degrees of freedom - a given number per space-time point. Their quantization is a subtle issue since approaches that are too naive lead to divergent results. In order to properly define a quantum field theory, one must also specify the integration measure of the fields in a path integral. One approach is to expand the path integral in powers of the coupling constant. The resulting Feynman diagrams are then regularized order by order in the coupling. This perturbative approach to field theory has led to impressive results in weakly interacting theories. Still, it is an asymptotic expansion, which is not entirely satisfactory even for QFTs with weak interactions, let alone strongly-coupled theories like QCD.

Since perturbative theories are not so useful in certain cases, people started developing non-perturbative approaches to QFTs, with lattice field theory (LFT) being one of the most prominent ones. LFT is formulated on a discretized Euclidean spacetime. It certainly is not an approximation to continuum field theory. It instead defines a theory that is undefined directly in the continuum. A significant advantage of this approach is that it can be simulated on a computer using the same techniques one uses to simulate problems in statistical mechanics.

QFTs always present us with notorious integrals that are hard to solve analytically. So, people often resort to *Monte Carlo* (MC) techniques to simulate such theories (approximating expectation values). However, MC simulation, too, has its setbacks. One can apply MC confidently only when the probability distribution is a real function. For example, in QCD with a finite chemical potential or with a theta term, the action of the field becomes a complex number, thereby making the Boltzmann weight also complex. In these cases we encounter a *complex action problem* or *sign problem*. An approach to tackle this problem is *complex Langevin simulation*, where we extend the concept of stochastic quantization of ordinary field theories with real actions to the case of complex actions. Such an extension leads to the complexification of field configurations.

This report is organized as follows. In Sec. 2 we discuss Langevin dynamics of real and complex fields (that is, fields under the influence of a complex potential). In Sec. 3, we discuss QFT in zero dimensions constructively. That is, we discuss how to build QFTs (both bosonic and fermionic) in zero dimensions. We close the section with a brief discussion on the construction of supersymmetric QFTs in zero dimensions. In Sec. 4, we talk about how one can explore SUSY breaking in zero-dimensional supersymmetric QFTs both by analytical method and by complex Langevin (CL) simulation. In Sec. 5, we check for the severity of the sign problem in two different zero-dimensional supersymmetric QFTs. In Sec. 6, we conclude and attempt to give an insight into our plans for expanding this project.

2 Complex Langevin Dynamics

2.1 Real Langevin Equation

We are interested in finding the expectation values of the observables $O(\phi)$ with the action being $S(\phi)$. We have

$$\langle O(\phi) \rangle = \frac{1}{Z} \int d\phi f(\phi) \exp(-S[\phi]),$$

where

$$Z = \int d\phi \exp(-S[\phi]).$$

Thus, as in plain MC, we want a method to sample the field configurations distributed with probability distribution function, $\text{pdf} \approx \exp(-S)$. One possible approach is the Langevin dynamics.

For simplicity, we consider a system with only one degree of freedom, say ϕ . Then the action is $S(\phi)$. The formulation can straightforwardly be generalized to systems with many degrees of freedom. The Langevin equation [1] is given by

$$\frac{d\phi(t)}{dt} = -\frac{dS(\phi)}{d\phi} + \eta(t). \quad (1)$$

Here the action is taken to be real. The gradient of $S(\phi)$ in the above equation is called the *drift term*, t is the fictitious time or the *Langevin time*, and $\eta(t)$ is a Gaussian noise with the following properties

$$\langle \eta(t) \rangle_\eta = 0 \quad \text{and} \quad \langle \eta(t)\eta(t') \rangle_\eta = 2\delta(t - t'). \quad (2)$$

Eq. (1) is a stochastic differential equation, so it is quite obvious that the solution to the equation, $\phi(t)$ subject to the noise, ought to have a probability distribution $p(\phi, t)$ at anytime t . The fundamental idea of Langevin dynamics is that, in the $t \rightarrow \infty$ limit $p(\phi) \propto \exp(-S(\phi))$, at least for $\forall S \in \mathbb{R}$. Thus,

$$\lim_{t \rightarrow \infty} \langle O(\phi_\eta(t)) \rangle_\eta = \lim_{t \rightarrow \infty} \int d\phi p(\phi, t) O(\phi) \propto \int d\phi O(\phi) \exp(-S(\phi)).$$

2.1.1 Fokker-Planck Equation

We aim to derive an evolution equation for $\langle O(\phi_\eta(t)) \rangle_\eta$. We see that

$$\frac{\langle dO(\phi_\eta(t)) \rangle_\eta}{dt} = \int d\phi O(\phi) \partial_t p(\phi, t). \quad (3)$$

Now from Eq. (1) we have

$$d\phi(t) = -S'(\phi)dt + dW(t) \quad \text{where} \quad dW(t) = \int_t^{t+dt} \eta(\tau) d\tau. \quad (4)$$

Thus, we see that

$$\begin{aligned} \langle dW(t) \rangle_\eta &= 0 & \langle dW^2(t) \rangle_\eta &= \int_t^{t+dt} \int_t^{t+dt} \langle \eta(\tau)\eta(\tau') \rangle_\eta d\tau d\tau' \\ & & &= 2 \int_t^{t+dt} \int_t^{t+dt} \delta(\tau - \tau') d\tau d\tau' = 2dt. \end{aligned}$$

We see that, in order to have a Taylor expansion up to first order in dt in Eq. (1), we have to expand $dO(\phi_\eta(t))$ up to second order in $d\phi$. So, we make use of Eqs. (1), (2), and (4), and find that

$$\begin{aligned} \langle dO(\phi(t)) \rangle_\eta &= \left\langle O'(\phi)d\phi + \frac{1}{2}O''(\phi)(d\phi)^2 \right\rangle_\eta \\ &= \left\langle -O'(\phi)S'(\phi)dt + O'(\phi)dW(t) + \frac{1}{2}O''(\phi)dW^2(t) - O''(\phi)dW(t)dt \right\rangle_\eta \\ &= \langle -O'(\phi)S'(\phi) + O''(\phi) \rangle_\eta dt \\ &\implies \frac{\langle dO(\phi_\eta(t)) \rangle_\eta}{dt} = \langle -O'(\phi)S'(\phi) + O''(\phi) \rangle_\eta. \end{aligned} \quad (5)$$

Now, from the definition of the expectation value, we find that

$$\begin{aligned}
\langle -O'(\phi)S'(\phi) + O''(\phi) \rangle_\eta &= \int d\phi \left\{ -O'(\phi)S'(\phi) + O''(\phi)p(\phi, t) \right\} \\
&= \int d\phi \left\{ O(\phi)\partial_\phi^2 p(\phi, t) + O(\phi)\partial_\phi S'(\phi)p(\phi, t) \right. \\
&\quad \left. - \partial_\phi \left[O(\phi)p(\phi, t)S'(\phi) + O'(\phi)p(\phi, t) - O(\phi)p(\phi, t) \right] \right\} \\
&= \int d\phi O(\phi) \{ \partial_\phi^2 + \partial_\phi S'(\phi) \} p(\phi, t).
\end{aligned}$$

The last line has been obtained by performing integration by parts and using vanishing boundary conditions at infinity.

Now, comparing the result obtained above with Eq. (3) we get

$$\partial_t p(\phi, t) = (\partial_\phi^2 + \partial_\phi S'(\phi))p(\phi, t) = -H_{\text{FP}}p(\phi, t). \quad (6)$$

Equation (6) is called the *Fokker-Planck* equation, and H_{FP} is called the *Fokker-Planck Hamiltonian*.

Now, if we do the following similarity transformation

$$\tilde{p}(\phi, t) = e^{\frac{1}{2}S(\phi)}p(\phi, t),$$

the Fokker-Planck equation becomes

$$\partial_t \tilde{p}(\phi, t) = -\tilde{H}_{\text{FP}}\tilde{p}(\phi, t),$$

where

$$\begin{aligned}
\tilde{H}_{\text{FP}} &= e^{\frac{1}{2}S(\phi)}H_{\text{FP}}e^{-\frac{1}{2}S(\phi)} \\
&= \left(-\partial_\phi + \frac{1}{2}S'(\phi) \right) \left(\partial_\phi + \frac{1}{2}S'(\phi) \right) \\
&= A^\dagger A,
\end{aligned}$$

with the matrix A defined as

$$A \equiv \left(\partial_\phi + \frac{1}{2}S'(\phi) \right).$$

We have

$$\tilde{H}_{\text{FP}}^\dagger = \tilde{H}_{\text{FP}}$$

and

$$\langle \alpha | \tilde{H}_{\text{FP}}(\alpha) \rangle = \langle A(\alpha) | A(\alpha) \rangle \geq 0 \quad \forall \alpha \in \mathcal{M} \quad \text{and} \quad \mathcal{M} \text{ is } L^2.$$

Thus we find that $\tilde{H}_{\text{FP}}$ is both Hermitian and non-negative. It possesses a complete eigenbasis with non-negative eigenvalues. Also, we observe that

$$\begin{aligned}
&\tilde{H}_{\text{FP}}\psi_0 = 0 \\
\implies &\left(\partial_\phi + \frac{1}{2}S'(\phi) \right) \psi_0(\phi) = 0 \\
\implies &\psi_0(\phi) \propto \exp(-S(\phi)/2).
\end{aligned}$$

So, the eigenfunction with lowest eigenvalue ($= 0$) is proportional to $\exp(-S(\phi)/2)$. Now, we can expand $\tilde{p}(\phi, t)$ as follows

$$\begin{aligned} p(\tilde{\phi}, t) &= \exp(-\tilde{H}_F P t) \left[c_0 \exp(-S(\phi)/2) + \sum_n c_n \psi_n(\phi, t=0) \right] \\ &= c_0 \exp(-S(\phi)/2) + \sum_n c_n \psi_n(\phi, t=0) \exp(-E_n t) \quad E_n > 0 \quad \forall n \in \mathbb{N}. \end{aligned}$$

So, in equilibrium, we would have

$$p_{eq}(\phi) = \lim_{t \rightarrow \infty} p(\phi, t) = \exp(-S(\phi)/2) \lim_{t \rightarrow \infty} \tilde{p}(\phi, t) \propto \exp(-S(\phi)).$$

This fact, together with the ergodicity hypothesis, makes the Langevin equation, Eq. (1) a convenient tool for producing sets of solutions distributed according to $\exp(-S(\phi))$.

However, when t is greater than a certain time, t_{th} needed for the thermalization of the system, the solutions fluctuate around the thermal equilibrium, and we can therefore calculate the average [1] from one solution over a large time interval T

$$\langle O(\phi_\eta(t)) \rangle \approx \frac{1}{T} \int_{t_{th}}^{t_{th}+T} d\tau O(\phi_\eta(\tau)) \approx \langle O(\phi) \rangle_s.$$

2.2 Discretized Langevin Dynamics

The Langevin equation, Eq. (1) in discrete form becomes

$$\phi_{n+1} = \phi_n - S'(\phi_n) \Delta t + \eta_n \Delta t. \quad (7)$$

Here η_n is a discrete sequence of noise having a Gaussian signature. We have

$$\begin{aligned} \langle \eta(t) \eta(t') \rangle &= 2\delta(t-t') \longrightarrow \langle \eta_n \eta_m \rangle = \alpha \delta_{nm} \\ \int \langle \eta(t) \eta(t') \rangle dt &= 2 \longrightarrow \Delta t \sum_n \langle \eta_n \eta_m \rangle = 2 \\ \implies \eta_n &= \sqrt{\alpha} \beta_n = \sqrt{\frac{2}{\Delta t}} \beta_n \quad \langle \beta_m \beta_n \rangle = \delta_{mn}. \end{aligned}$$

So Eq. (6) becomes

$$\phi_{n+1} = \phi_n - S'(\phi_n) \Delta t + \sqrt{2\Delta t} \beta_n. \quad (8)$$

2.3 Complex Langevin Equation

When the action $S(\phi)$ becomes complex (assuming holomorphic complex function), the function $p_c(\phi, t)$ as obtained above in general, becomes a complex function. It hence loses its name as the probability distribution. Also, the similarity transformed Hamiltonian loses non-negativity and Hermiticity in general, so one cannot guarantee about convergence of the limit $\lim_{t \rightarrow \infty} p(\phi, t)$ to $\exp(-S(\phi))$.

One way to approach this problem is to complexify the field variables. So, we write $\phi = x + iy$. The expectation values of observables become

$$\langle O(\phi_\eta(t)) \rangle_\eta = \int dx dy p(x, y, t) O(x + iy),$$

where $p(x, y, t)$ is a real function of x and y and can be interpreted as a probability distribution.

The Fokker-Planck Hamiltonian becomes

$$H_{\text{FP}} = -\partial_x^2 - \partial_x \text{Re}(S'(\phi)) - \partial_y \text{Im}(S'(\phi)).$$

The discretized Langevin equation becomes

$$\begin{aligned} x_{n+1} &= x_n - K_1(x_n, y_n)\Delta t + \sqrt{2\Delta t}\beta_n, \\ y_{n+1} &= y_n - K_2(x_n, y_n)\Delta t, \\ K_1(x_n, y_n) &= \text{Re}(S'(\phi_n)), \\ K_2(x_n, y_n) &= \text{Im}(S'(\phi_n)). \end{aligned} \tag{9}$$

3 Zero-Dimensional QFT

3.1 Choice of a Manifold $\mathcal{M}$

The process of building a quantum field theory starts with choosing a manifold $\mathcal{M}$ of dimension d on which certain objects called *fields* can be defined. For most but all QFTs, the manifolds are viewed as Riemannian with a smooth metric. When the metric is positive definite, we call it an *Euclidean QFT*. For many physical applications, we will consider manifolds with metrics with signature $\neq d$. One such example is Minkowski space with the Minkowski metric.

3.2 Choice of Objects (Fields) and Action

The next objective is to define fields on the chosen manifold. In QFTs, one aims to integrate over the space parametrizing the fields. These integrals are called *path integrals*. For example, we may consider a principal bundle with a connection over $\mathcal{M}$. In physics terminology, the choice of the connection is called “picking a gauge field.” We may also be considering sections of a vector bundle over $\mathcal{M}$. These fields are sometimes called matter fields. Quantum field theories associated with connections and sections of associated vector bundles are called *quantum gauge theories*.

We must define a measure of the field space when it is about integration. There is a usual choice for that in most cases. In the case of Euclidean signature, the measure is weighted by $\exp(-S)$. In the case of Minkowski space, it is $\exp(-iS)$. Here S is a functional defined on the space of fields.

The case of $d = 0$, means $\mathcal{M}$ is a point. In this case, QFT is equivalent to carrying out some finite-dimensional integrals. We use this simple case to set up the basic ingredients of quantum field theories and introduce fermionic fields and supersymmetry. In this context, we can discuss rigorously the important notions of localization and deformation invariance that often arise in supersymmetric quantum field theories.

3.3 Super Vector Spaces

We will be discussing field theory with supersymmetry. That means the QFT will involve both bosonic and fermionic variables. Now, to consider any theory involving fermions, one needs the notion of Grassmann variables [2, 3] and their algebra.

We start with a $\mathbb{Z}_2$ -graded vector space, a vector space of the form $V = V_0 \oplus V_1$. The vector spaces can be assumed to be defined over the fields $\mathbb{R}$ or $\mathbb{C}$.

A vector v is called *homogeneous* if it belongs purely to V_0 or V_1 . A notion called parity ($|\cdot|$) can be defined for the homogeneous vectors as

$$|v| = \begin{cases} 0 & v \in V_0 \\ 1 & v \in V_1 \end{cases}$$

The elements with parity 0 are often called *even* while the others are called *odd*. In the context of physics, they can be called *bosons* and *fermions*, respectively.

If $\dim(V_0) = p$ and $\dim(V_1) = q$ then dimension of V is written as $\dim(V) = p|q (= p + q)$. For example, if $V_0 = \mathbb{R}^p$ and $V_1 = \mathbb{R}^q$, we have $V = \mathbb{R}^{p|q}$.

Now, what makes V a super vector space is an unusual choice of symmetry operation. Before

that, we will see what direct product of two $\mathbb{Z}_2$ -graded vector spaces results in

$$\begin{aligned} V \otimes W &= A_0 \oplus A_1, \\ A_0 &= (V_0 \otimes W_0) \oplus (V_1 \otimes W_1), \\ A_1 &= (V_0 \otimes W_1) \oplus (V_1 \otimes W_0). \end{aligned}$$

The direct product respects $\mathbb{Z}_2$ grading.

For super vector spaces, the exchange operation is given by

$$s : V \otimes W \rightarrow W \otimes V, \quad s(v \otimes w) = (-1)^{|v||w|}(w \otimes v). \quad (10)$$

Now, to make V a *superalgebra*, we define the following composition (multiplication) on the V

$$(\cdot) : V \times V \rightarrow V,$$

that preserves $\mathbb{Z}_2$ grading in a sense that $|v \cdot w| = |v| + |w|$. The $\cdot$ symbol can be avoided. A superalgebra is said to be supercommutative if

$$vw = (-1)^{|v||w|}wv. \quad (11)$$

For example, if $V = \mathbb{R}^{p|q}$ we have

$$\begin{aligned} X_i X_j &= X_j X_i, \\ X_i \psi_j &= \psi_j X_i, \\ \psi_i \psi_j &= -\psi_j \psi_i, \end{aligned} \quad (12)$$

where X_i is a bosonic variable while ψ_i is a fermionic variable.

Thus, our superalgebra must contain at least two fermionic variables if it has any non-trivial content (beyond a usual algebra).

Now we will define differentiation and integration on V .

Differentiation

The derivative operator defined on supercommutative superalgebra V would obey the following rule

$$D(vw) = D(v)w + (-1)^{|v||w|}vD(w) \quad \forall v, w \in V. \quad (13)$$

If $V = \mathbb{R}^{p|q}$ then we have both even (bosonic) and odd (fermionic) derivatives, which are $\frac{\partial}{\partial X_i}$ and $\frac{\partial}{\partial \psi_i}$, respectively. We have

$$\begin{aligned} \partial_{X_i} X_j &= \delta_i^j, & \partial_{X_i} \psi_j &= 0, \\ \partial_{\psi_i} X_j &= 0, & \partial_{\psi_i} \psi_j &= \delta_i^j. \end{aligned} \quad (14)$$

Also

$$\partial_{\psi_i} \psi_j \psi_k = \delta_i^j \psi_k - \psi_j \delta_i^k.$$

A smooth vector field on $\mathbb{R}^{p|q}$ is the following derivative

$$\xi(\phi, \psi) = X(\phi, \psi)\partial_\phi + \chi(\phi, \psi)\partial_\psi.$$

If X is even and χ is odd, then ξ is even, and if X is odd and χ is even then ξ is odd.

Integration

Any function of a single fermionic variable is $f + \rho\psi$. We demand the integration measure to be translationally invariant [2]. Then for $\psi' = \psi + \rho$ we have

$$\begin{aligned} \int \psi' d\psi' &= \int (\psi + \rho) d\psi = \int \psi d\psi + \rho \int d\psi \\ \implies \int d\psi &= 0 \quad \int \psi d\psi = 1 \quad \text{the integration measure is normalized} \end{aligned} \tag{15}$$

This also means that for any fermionic function $F(\psi)$ as we must have $F(\psi) = a + b\psi$

$$\int \partial_\psi F(\psi) d\psi = 0. \tag{16}$$

The rules mentioned above are called *rules of Berezin integration*.

For n such fermionic variables, we have

$$\begin{aligned} \int \psi_1 \psi_2 \cdots \psi_n d^n \psi &= 1, \\ \int \psi_{a_1} \psi_{a_2} \cdots \psi_{a_n} d^n \psi &= \epsilon_{a_1 a_2 a_3 \cdots a_n}. \end{aligned}$$

If we consider a change of variables by some linear transformation $T \in GL_n$. We will have

$$\begin{aligned} \chi_i &= T_i^k \psi_k \quad \forall k, i \\ \int \chi_{a_1} \chi_{a_2} \cdots \chi_{a_n} d^n \psi &= T_{a_1}^{b_1} T_{a_2}^{b_2} \cdots T_{a_n}^{b_n} \int \psi_{a_1} \psi_{a_2} \cdots \psi_{a_n} d^n \psi \\ &= \det(T) \epsilon_{a_1 a_2 a_3 \cdots a_n} \\ &= \int \chi_{a_1} \chi_{a_2} \cdots \chi_{a_n} d^n \chi \\ \implies d^n \chi &= \frac{1}{\det(T)} d^n \psi. \end{aligned}$$

For a similar transformation of bosonic variables, we have

$$d^n X = \det(T) d^n \phi.$$

3.4 Purely Bosonic Theory

In the case of $d = 0$, the fields, $\phi : \mathcal{M} \mapsto \mathbb{R}$ are real (or complex) variables. The action $S[\phi]$ is just a function of ϕ , and the partition function is given by

$$Z = \int d\phi \exp(-S[\phi]).$$

The correlation functions are given by

$$\langle f(\phi) \rangle = \frac{1}{Z} \int d\phi f(\phi) \exp(-S[\phi]).$$

One can also deform the action as shown below

$$S \longrightarrow S + \sum_i \alpha_i f_i(\phi)_i.$$

Then

$$\langle f(\phi) \rangle = - \left. \frac{\partial \log Z}{\partial \alpha_i} \right|_{\alpha_i=0}.$$

For bosonic fields, the variables ϕ (field) follow the algebra of the field of reals (complex numbers).

For N such fields, the space of field configurations $\mathcal{C}$ becomes

$$\mathcal{C} \cong \{\phi_1, \phi_2, \dots, \phi_N | \phi_i \in \mathbb{R} \ \forall \ i\} = \mathbb{R}^N.$$

The path-integral measure becomes the standard (Lebesgue) measure in $\mathbb{R}^N$

$$D[\phi] \rightarrow d^N \phi.$$

The partition function becomes

$$\mathcal{Z} = \int_{\mathbb{R}^n} d^n \phi \exp(-S(\phi_1, \phi_2, \dots, \phi_n)).$$

The action for a purely bosonic theory in general looks like

$$S(\phi) = \frac{1}{2} \phi^i M_{ij} \phi^j + V(\phi_1, \phi_2, \dots, \phi_n),$$

where M_{ij} is positive definite, and invertible. For a free theory, the potential term is zero. For a free theory, one has

$$\mathcal{Z}_0 = \int_{\mathbb{R}^n} \exp\left(-\frac{1}{2} \phi^i M_{ij} \phi^j\right) d^n \phi.$$

Given M_{ij} is positive definite in a complex vector space, M_{ij} is also Hermitian. So, M_{ij} is diagonalizable. Then one can have

$$\begin{aligned} \mathcal{M} &\rightarrow U^{-1} M U = \mathcal{D}, \\ (\phi_1, \phi_2, \dots, \phi_n) U &= (X_1, X_2, \dots, X_n), \\ \phi^i M_{ij} \phi^j &\rightarrow \lambda_i X^i X^i. \end{aligned}$$

Then we have the partition function for the free theory part of the action

$$\begin{aligned} \mathcal{Z}_0 &= \int_{\mathbb{R}^n} \exp\left(-\frac{1}{2} \lambda_i X^i X^i\right) d^n X \\ &= \frac{(\sqrt{2\pi})^n}{\sqrt{\prod_i \lambda_i}} \\ &= \frac{(\sqrt{2\pi})^n}{\sqrt{\det(M)}}. \end{aligned}$$

We can also hope to perturbatively expand the partition function around the classical limit given by $\hbar \rightarrow 0^+$. In this limit, the weight $\exp(-S(\phi)/\hbar)$ only takes contributions from near the classical minima. Given that the minimum is unique and isolated, we would have the Hessian of S positive definite at that point. For clarity, we can write

$$S(\phi) = S(\phi_0) + \partial_{\phi_1} \partial_{\phi_j} S(\phi_0) \delta \phi \delta \phi_j + O(\delta \phi^3).$$

We also have

$$\mathcal{Z} \approx (2\pi\hbar)^{n/2} \frac{e^{-S(\phi_0)/\hbar}}{\sqrt{\det(\partial_{\phi_1} \partial_{\phi_j} S(\phi_0))}} (1 + A_1\hbar + A_2\hbar^2 + \dots).$$

It can be obtained using the method of steepest descent. We see that the leading term in the asymptotic series of $\mathcal{Z}$ is $\mathcal{Z}_0$.

3.5 Purely Fermionic Theory

Even for a $d = 0$ QFT of fermions, the action as a mathematical object has to be a bosonic variable. That is why we have to consider at least two fermions in our theory. Then the simplest action would be of the form

$$S(\psi_1, \psi_2) = A\psi_1\psi_2.$$

The partition function is given by

$$\mathcal{Z} = \int e^{-S/\hbar} d^2\psi = \int \exp(-A\psi_1\psi_2/\hbar) d^2\psi = \int \int \left(1 - \frac{A}{\hbar}\psi_1\psi_2\right) d\psi_1 d\psi_2 = -\frac{A}{\hbar}.$$

This is the Gaussian action of the fermionic theory.

Using the same reasoning as above, one can understand that the action for fermionic theory must involve an even number of Grassmann variables. For a free fermionic theory, we can consider $2m$ fermions and an anti-symmetric matrix A_{ab} . Then the action has the form

$$S(\psi) = \frac{1}{2} A_{ij} \psi^i \psi^j.$$

The partition function is

$$\begin{aligned} \mathcal{Z} &= \int \exp(-S(\psi)/\hbar) d^{2m}\psi \\ &= \sum_{n=0}^m \frac{(-1)^n}{(2\hbar)^n n!} \int (A_{ij} \psi^i \psi^j d^{2m}\psi) \\ &= \frac{(-1)^m}{(2\hbar)^m m!} \int A_{a_1 a_2} A_{a_2 a_3} \cdots A_{a_{2m-1} a_{2m}} \psi^{a_1} \psi^{a_2} \cdots \psi^{a_{2m}} d^{2m}\psi \\ &= \frac{(-1)^m}{(2\hbar)^m m!} \epsilon^{a_1 a_2 \cdots a_{2m}} A_{a_1 a_2} A_{a_2 a_3} \cdots A_{a_{2m-1} a_{2m}} \\ &= \left(-\frac{1}{\hbar}\right)^m \text{Pfaff}(A). \end{aligned}$$

For an anti-symmetric matrix, we have

$$\text{Pfaff}(A) = \pm \sqrt{\det(A)}.$$

So, the partition function for a fermionic theory is proportional to $\sqrt{\det(A)}$, while for a bosonic theory, it is proportional to the reciprocal of $\det(M)$.

3.6 Zero-Dimensional Supersymmetric QFT

Now we discuss supersymmetric field theory in zero dimensions. A supersymmetric theory involves both bosons and fermions. The simplest of such theories contains one boson and two fermions (for reasons discussed in the last section), and the action in generic form looks like

$$S(\phi, \psi_1, \psi_2) = f(\phi) + V(\phi)\psi_1\psi_2.$$

Here, $f(\phi)$ and $V(\phi)$ are smooth functions of ϕ . To have a more clear picture, we first compute the path integral

$$\begin{aligned}\mathcal{Z} &= \int \exp(-f(\phi)/\hbar) \exp\left(-\frac{V(\phi)}{\hbar}\psi_1\psi_2\right) d\phi d\psi_1 d\psi_2 \\ &= -\frac{1}{\hbar} \int V(\phi) \exp(-f(\phi)/\hbar) d\phi \\ &= \int \exp(-\tilde{S}(\phi)/\hbar) d\phi \quad \tilde{S}(\phi) = f(\phi) - \hbar \log(-V(\phi)/\hbar).\end{aligned}$$

So the supersymmetric theory can be reduced to a purely bosonic theory with an effective action. Also, in the theory above $f(\phi)$ describes bosonic interactions, if it is a polynomial, then the individual monomials describe the interactions between bosons and can be represented as vertices in Feynman diagrams. $V(\phi)$ as a non-constant function describes the couplings between bosonic and fermionic fields.

We change the variables in the following way

$$\begin{aligned}\psi &= \psi_1 + i\psi_2, & \bar{\psi} &= \psi_1 - i\psi_2, \\ \{\psi, \bar{\psi}\} &= 0, & \psi^2 &= \bar{\psi}^2 = 0 & \because \{\psi_i, \psi_j\} &= 0 \quad \forall i, j.\end{aligned}$$

We also choose $f(\phi)$ and $V(\phi)$ such that the action becomes

$$S(\phi, \psi, \bar{\psi}) = \frac{1}{2}(\partial_\phi W)^2 - \bar{\psi}\psi\partial_\phi^2 W. \quad (17)$$

Here $W(\phi)$ is a smooth function of ϕ .

Supersymmetry

A symmetry transformation of a field theory is a transformation that leaves the action invariant. For a theory that contains both bosons and fermions, if the symmetry transformation changes the bosonic fields to fermionic fields and vice versa, we call it a supersymmetry transformation.

For the $d = 0$ theory we described above, let us consider the following fermionic vector fields [2] (they are called fermionic as per the definition of odd vector fields discussed in the previous section)

$$\begin{aligned}Q &= \psi\partial_\phi + \partial_\phi W\partial_{\bar{\psi}}, \\ Q^\dagger &= \bar{\psi}\partial_\phi - \partial_\phi W\partial_\psi.\end{aligned} \quad (18)$$

Now we observe that

$$\begin{aligned}Q(\phi) &= \psi, & Q^\dagger(\phi) &= \bar{\psi}, \\ Q(\psi) &= 0, & Q^\dagger(\psi) &= -\partial_\phi W, \\ Q(\bar{\psi}) &= \partial_\phi W, & Q^\dagger(\bar{\psi}) &= 0.\end{aligned}$$

Also

$$Q(S) = \psi \partial_\phi(S) = \psi \partial_\phi W \partial_\phi^2 W - \psi \partial_\phi W \partial_\phi^2 W = 0 = Q^\dagger(S).$$

So, by definition of supersymmetry, we can say that Q and $Q^\dagger$ generate supersymmetry transformations.

Partition Function

The partition function for the theory discussed above is

$$\mathcal{Z} = \frac{1}{\hbar} \int \partial_\phi^2 W \exp(-(\partial_\phi W)/\hbar) d\phi.$$

We normalize the integration measure such that it becomes

$$\mathcal{Z} = \frac{1}{\sqrt{2\pi}} \int \partial_\phi^2 W \exp(-(\partial_\phi W)) d\phi \quad (\hbar = 1).$$

There are two very different ways to evaluate $\mathcal{Z}$, although both use the *principle of localization*.

First method: In this method we scale $\partial_\phi W$ by some $\alpha \in \mathbb{R}_{>0}$. Then

$$S_\alpha(\phi, \psi, \bar{\psi}) = \frac{1}{2} \alpha^2 (\partial_\phi W)^2 - \alpha \bar{\psi} \psi \partial_\phi^2 W.$$

The supersymmetric charges become

$$\begin{aligned} Q_\alpha &= \psi \partial_\phi + \alpha \partial_\phi W \partial_{\bar{\psi}}, \\ Q_\alpha^\dagger &= \bar{\psi} \partial_\phi - \alpha \partial_\phi W \partial_\psi. \end{aligned}$$

Now we define

$$\mathcal{I}_\alpha = \frac{1}{\sqrt{2\pi}} \int e^{-S_\alpha} d\phi d\psi d\bar{\psi}.$$

We have $\mathcal{I}_1 = \mathcal{Z}$. Now

$$\begin{aligned} \mathcal{I}'_\alpha &= \frac{1}{\sqrt{2\pi}} \frac{d}{d\alpha} \int e^{-S_\alpha} d\phi d\psi d\bar{\psi} \\ &= -\frac{1}{\sqrt{2\pi}} \int (\alpha (\partial_\phi W)^2 - \bar{\psi} \psi \partial_\phi^2 W) e^{-S_\alpha} d\phi d\psi d\bar{\psi} \\ &= \frac{1}{\sqrt{2\pi}} \int Q_\alpha^\dagger (\psi \partial_\phi W e^{-S_\alpha}) d\phi d\psi d\bar{\psi} \\ &\quad \because Q_\alpha^\dagger (\psi \partial_\phi W) = -(\alpha (\partial_\phi W)^2 - \bar{\psi} \psi \partial_\phi^2 W) \text{ and } Q_\alpha^\dagger (S_\alpha) = 0 \\ &= \frac{1}{\sqrt{2\pi}} \int \bar{\psi} \psi \partial_\phi (W'(\phi) \exp(-S_\alpha)) d\phi d\psi d\bar{\psi} \\ &= \frac{1}{\sqrt{2\pi}} \int \bar{\psi} \psi \partial_\phi \left(W'(\phi) \exp(-\alpha^2 W'^2/2) [1 + \alpha \bar{\psi} \psi W''(\phi) + \dots] \right) d\phi d\psi d\bar{\psi} \\ &= \frac{1}{\sqrt{2\pi}} \int \partial_\phi (W'(\phi) \exp(-\alpha^2 W'^2/2)) d\phi. \end{aligned}$$

Now, the integrand is a total derivative with respect to ϕ . We can carry out the ϕ -integral first this time. Now $W(\phi)$, as a real superpotential is often chosen to be a real polynomial. In those cases we have

$$\lim_{|\phi| \rightarrow \infty} W'(\phi) \exp(-W'^2/2) = 0,$$

implying

$$\frac{d\mathcal{I}}{d\alpha_\alpha} = 0.$$

So, we can set

$$\mathcal{Z} = \mathcal{I}_1 = \lim_{\alpha \rightarrow \infty} \mathcal{I}_\alpha.$$

Now for large α , only the points where $W'(\phi) = 0$ contribute to the integral; the other regions are suppressed by $\exp(-\alpha^2 W'^2/2)$ and will contribute nothing.

So, if $W'(\phi) \neq 0 \forall \phi \in \mathbb{R}$ then certainly $\mathcal{Z} = 0$, which leads to SUSY breaking, which we would discuss in brief later.

Second Method: We would show that this method leads to the same conclusion as above, and then we would use localization to evaluate the integral.

So, we first consider $A = \{\phi | W'(\phi) = 0\}$. Let $B = \mathbb{R} - A$. We can then make the following change of variables in B

$$\begin{aligned}\chi &= \phi - \frac{\bar{\psi}\psi}{W'}, \\ \psi_1 &= \sqrt{W'}\psi, \\ \psi_2 &= \bar{\psi}.\end{aligned}$$

The Jacobian matrix of the transformation is

$$\begin{pmatrix} 1 + \frac{\bar{\psi}\psi W''}{W'^2} & +\frac{\bar{\psi}}{W'} & \frac{\psi}{W'} \\ \frac{W''\psi}{2\sqrt{W'}} & \sqrt{W'} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

We also have

$$\det(J) = \sqrt{W'} + \frac{3}{2} \frac{\bar{\psi}\psi W''}{W'^{\frac{3}{2}}} = \dot{W}(\chi) \quad \because \phi = \chi + \frac{\psi_2\psi_1}{\sqrt{W'^3}}.$$

We also see that

$$\begin{aligned}W'(\phi) &= \dot{W}(\chi) \left(1 + \frac{\bar{\psi}\psi}{W'^2}\right), \\ W''(\phi)\bar{\psi}\psi &= \dot{W}(\chi)\bar{\psi}\psi, \\ S(\phi, \bar{\psi}, \psi) &= \frac{1}{2}(\dot{W})^2 = \tilde{S}(\chi, 0, 0).\end{aligned}$$

Then

$$\mathcal{Z} = \int \exp(-S) d\phi d\psi d\bar{\psi} = \int \exp(-\tilde{S}(\chi, 0, 0)) \sqrt{\dot{W}} d\chi d\psi_1 d\psi_2 = 0.$$

As before, the non-vanishing contribution to the integral comes only from the neighborhood of points where $W'(\phi) = 0$.

Localization: The principle of localization [4] is as simple as localizing the domain of integral to neighborhoods of zeros of $W'(\phi)$.

We assume that $W'(\phi)$ has unique isolated zeros. Let x^* be an isolated zero. Then in some neighborhood of x^* , we have

$$W(\phi) \approx W(x^*) + \frac{1}{2}W''(x^*)(\phi - x^*)^2,$$

$$S(\phi) \approx \frac{\lambda^2}{2}(x - \phi)^2 - \lambda\bar{\psi}\psi, \quad \lambda = W''(x^*).$$

Then the contribution of this neighborhood to the integral is

$$\frac{1}{\sqrt{2\pi}} \int \exp(-\lambda(\phi - x^*)^2/2)(-1 + \lambda\bar{\psi}\psi)d\phi d\psi d\bar{\psi} = \frac{\lambda}{|\lambda|} = \text{sign}(W''(x^*)),$$

giving

$$\mathcal{Z} = \sum_{x^*|W'(x^*)=0} \text{sign}(W''(x^*)) = \pm 1.$$

So, in the supersymmetric field theory discussed above we have $\mathcal{Z} \in \{-1, 0, 1\}$.

4 SUSY Breaking in Zero-dimensional Field Theories

We consider a $d = 0$ supersymmetric QFT with the action [5] given by

$$S = \frac{1}{2}B^2 + iBW' + \bar{\psi}\psi W''.$$

Here B is an auxiliary field. (It is a field that appears in the Lagrangian, which is the action in a quadratic term with no kinetic term. It is a field whose equation of motion admits a single solution.)

We define the following fermionic vector fields

$$\begin{aligned} Q &= \psi\partial_\phi - iB\partial_{\bar{\psi}}, \\ Q^\dagger &= -\bar{\psi}\partial_\phi - iB\partial_\psi. \end{aligned} \tag{19}$$

We can show that

$$\begin{aligned} Q(\phi) &= \psi, & Q^\dagger(\phi) &= -\bar{\psi}, \\ Q(\psi) &= 0, & Q^\dagger(\psi) &= -iB, \\ Q(\bar{\psi}) &= -iB, & Q^\dagger(\bar{\psi}) &= 0, \\ Q(B) &= 0, & Q^\dagger(B) &= 0. \end{aligned}$$

The auxiliary field can also be eliminated using its equation of motion, i.e.,

$$\frac{\partial S}{\partial B} = 0 \implies iB = W'.$$

It is clear that

$$\begin{aligned} Q(S) &= 0, \\ Q^\dagger(S) &= 0. \end{aligned}$$

From the definition of SUSY, we can say that Q and $Q^\dagger$ generate a SUSY in the theory mentioned above.

The partition function can be calculated as

$$\begin{aligned}
\mathcal{Z} &= \frac{1}{2\pi} \int dB d\phi d\psi d\bar{\psi} e^{-S} \\
&= \frac{1}{\sqrt{2\pi}} \int d\phi d\psi d\bar{\psi} \exp\left(-\frac{1}{2}W'^2 + W''\bar{\psi}\psi\right) \\
&\quad \left(\because \int \exp(-\alpha X^2 + \beta X) dX = \sqrt{\pi/\alpha} \exp\left(-\frac{\beta^2}{4\alpha}\right)\right) \\
&= -\frac{1}{\sqrt{2\pi}} \int d\phi W'' \exp(-W'^2/2).
\end{aligned} \tag{20}$$

As discussed before, it is possible that $\mathcal{Z} = 0$. The expectation value of any observable will have $\mathcal{Z}$ in the denominator. So, $\mathcal{Z} = 0$ might lead to indefinite values of expectation values of observables. This leads to SUSY breaking (which will be discussed in a bit more detail later).

The expectation value of the auxiliary field plays a crucial role in investigating SUSY breaking. We have

$$\begin{aligned}
\langle B \rangle &= \frac{1}{2\pi\mathcal{Z}} \int dB d\phi d\psi d\bar{\psi} B e^{-S} \\
&= \frac{i}{2\pi\mathcal{Z}} \int d\phi W' W'' \exp(-W'^2/2) \\
&= -\frac{i}{2\pi\mathcal{Z}} \int d\phi \partial_\phi \exp(-W'^2/2).
\end{aligned} \tag{21}$$

The integrand is a total derivative so the integral is zero. So, when SUSY is broken, $\langle B \rangle$ becomes indefinite (That is, $0/0$ form is obtained).

In order to overcome this difficulty, we can introduce an external field and then eventually take a limit where it goes to zero. We usually introduce some external field to detect the spontaneous breaking of ordinary symmetry so that the ground state degeneracy is lifted to specify a single broken ground state. We take the thermodynamic limit of the theory, and after that, the external field is turned off. The value of the corresponding order parameter then would tell us if spontaneous symmetry breaking happens in the model or not.

We can apply the above mentioned scheme to the same theory obtained from a $d = 1$ supersymmetric QFT by reducing dimensions.

The action [6] is a one-dimensional integral over a time circle of circumference β (one-point compactified interval of length β) in Euclidean space. It is

$$S = \int_0^\beta d\tau \left(\frac{1}{2} B^2 + iB(\dot{\phi} + W') + \bar{\psi}(\dot{\psi} + W''\psi) \right).$$

There are also periodic boundary conditions (PBCs) imposed on the fields

$$\begin{aligned}
\phi(\tau) &= \phi(\tau + \beta), \\
\psi(\tau) &= \psi(\tau + \beta), \\
\bar{\psi}(\tau) &= \bar{\psi}(\tau + \beta), \\
B(\tau) &= B(\tau + \beta).
\end{aligned}$$

With the PBCs, the following integral becomes the *Witten index* [7]

$$\frac{1}{2\pi} \int DB D\phi D\psi D\bar{\psi} e^{-S} = \text{Tr}((-1)^F \exp(-\beta\mathcal{H})).$$

The following operators are supercharges in the $d = 1$ QFT

$$\begin{aligned} Q(\phi) &= \psi, & Q^\dagger(\phi) &= -\bar{\psi}, \\ Q(\psi) &= 0, & Q^\dagger(\psi) &= -iB + 2\dot{\phi}, \\ Q(\bar{\psi}) &= -iB, & Q^\dagger(\bar{\psi}) &= 0, \\ Q(B) &= 0, & Q^\dagger(B) &= 2i\dot{\psi}. \end{aligned}$$

Also, one can observe that

$$\begin{aligned} \{Q, Q\} &= 0, \\ \{Q^\dagger, Q^\dagger\} &= 0, \\ \{Q^\dagger, Q\} &= 2\frac{\partial}{\partial\tau} = 2\mathcal{H}, \end{aligned}$$

where $\mathcal{H}$ is the Hamiltonian and

$$F = \bar{\psi}\psi = \text{The fermionic number operator.}$$

Then it is clear that F has two distinct eigenvalues, 1 and 0, respectively. Furthermore, the two eigenspaces can make the Hilbert space a $\mathbb{Z}$ -graded vector space. It can be written as

$$H = (\mathbb{R} \otimes |0\rangle) \oplus (\mathbb{R} \otimes |1\rangle),$$

where

$$H_B = (\mathbb{R} \otimes |0\rangle) \quad H_F = (\mathbb{R} \otimes |1\rangle),$$

and

$$\begin{aligned} Q(H_B) &\subset H_F, & Q^\dagger(H_B) &\subset H_F, \\ Q(H_F) &\subset H_B, & Q^\dagger(H_F) &\subset H_B. \end{aligned}$$

We see that

$$\mathcal{H}|\alpha\rangle = 0 \implies Q|\alpha\rangle = Q^\dagger|\alpha\rangle = 0.$$

From the expression for the Hamiltonian, one can make sure that it is semi-positive definite. This means that the ground state has zero energy. The equations above imply that a state is a ground state if and only if it is invariant under SUSY transformation.

One can define a new operator

$$Q^2 = (Q + Q^\dagger)^2 = \{Q, Q^\dagger\} = 2\mathcal{H}.$$

For excited states, Q^2 can be an isomorphism between H_B and H_F , which would imply that for $n > 0$

$$H_B^n \cong H_F^n.$$

With a bit of algebra, we can have the following multiplets of states

$$|b_{n+1}\rangle = \frac{1}{\sqrt{2E_n}}Q^\dagger|f_n\rangle \quad \text{and} \quad |f_{n+1}\rangle = \frac{1}{\sqrt{2E_n}}Q|b_n\rangle.$$

We can set $|b_0\rangle$ to be the ground state setting $\dim(H_F^0) = 0$.

Also

$$\text{Tr}((-1)^F \exp(-\beta\mathcal{H})) = \dim(H_B^0) - \dim(H_F^0).$$

When the ground state is not invariant under supersymmetry, there is *spontaneous supersymmetry breaking* in the model. That also means there is no ground state. This, in turn, implies

$$\text{Tr}((-1)^F \exp(-\beta\mathcal{H})) = 0.$$

As mentioned at the beginning of this section, the partition function, which is also the Witten index, becomes zero when SUSY breaks.

Lattice Discretization

Now we discretize this theory on a lattice with T sites. The action [5] becomes

$$S = \sum_{n=0}^{T-1} \left[\frac{1}{2} B(n)^2 + iB(n)(\phi(n+1) - \phi(n) + W') + \bar{\psi}(n)(\psi(n+1) - \psi(n) + W''\psi) \right].$$

If we consider a lattice with one site (which corresponds to one single instant of time in one-dimensional QFT leading to $d = 0$ QFT) we have

$$S = \frac{1}{2} B(0)^2 + iB(0)[\phi(1) - \phi(0) + W'] + \bar{\psi}(0)[\psi(1) - \psi(0) + W''\psi].$$

With PBCs mentioned previously, we have

$$S = \frac{1}{2} B^2 + iBW' + W''\bar{\psi}\psi. \quad (22)$$

We mentioned before that to study the SUSY breaking in a more convenient way, we have to modify the boundary conditions (BCs). We introduce the twisted BCs,

$$\begin{aligned} \phi(1) &= \phi(0), \\ \psi(1) &= e^{i\alpha}\psi(0), \\ \bar{\psi}(1) &= e^{-i\alpha}\bar{\psi}(0), \\ B(1) &= B(0). \end{aligned} \quad (23)$$

Here α is a regularization parameter.

The action takes the following form

$$S_\alpha = \frac{1}{2} B^2 + iBW' + \bar{\psi}(e^{i\alpha} - 1 + W'')\psi. \quad (24)$$

In the $\alpha \rightarrow 0$ limit, we recover our original S .

Also,

$$QS_\alpha = -iQ^\dagger S_\alpha = (e^{i\alpha} - 1)\bar{\psi}\psi.$$

Again in the same limit, SUSY is recovered. So, one can compute the expectation values in the lattice-regularized QFT, then take the $\alpha \rightarrow 0$ limit, and check for SUSY breaking. That is,

$$\lim_{\alpha \rightarrow 0} \langle B_\alpha \rangle = \begin{cases} 0 & \implies \text{SUSY is preserved} \\ \neq 0 & \implies \text{SUSY is broken} \end{cases}$$

The partition function after carrying out the Berezin integration and eliminating B takes the following form

$$\mathcal{Z}_\alpha = -\frac{1}{\sqrt{2\pi}} \int d\phi (\exp(i\alpha) - 1 + W'') \exp(-W'^2/2) = -\frac{1}{\sqrt{2\pi}} \int d\phi \exp(-S_{\text{eff}}), \quad (25)$$

where

$$\begin{aligned} S_{\text{eff}} &= \frac{1}{2}W'^2 - \ln(e^{i\alpha} - 1 + W''), \\ S'_{\text{eff}} &= W'W'' - \frac{W'''}{e^{i\alpha} - 1 + W''}. \end{aligned} \quad (26)$$

One also has

$$\langle B_\alpha \rangle = \frac{1}{\sqrt{2\pi}\mathcal{Z}_\alpha} \int d\phi (iW') \exp(-S_{\text{eff}}). \quad (27)$$

In this section, we briefly discussed SUSY breaking in zero-dimensional supersymmetric QFT. In the next section, we will simulate the $d = 0$ QFT with a particular class of $W(\phi)$ using the complex Langevin method and check the severity of the sign problem in these theories.

5 Severity of the Sign Problem in $d = 0$ QFT

The complex Langevin simulations have been run with the drift vector $-S'_{\text{eff}}$ of the action given in Eq. (25).

5.1 Real Double-well Potential

The superpotential $W(\phi)$ is such that

$$W'(\phi) = g(\mu^2 + \phi^2),$$

where g is a coupling constant and μ is a real constant.

We have

$$S'_{\text{eff}} = 2\mu g^2(\mu^2 + \phi^2) - \frac{2g}{e^{i\alpha} - 1 + 2g\phi}.$$

We simulate the theory using Langevin equations, Eq. (8).

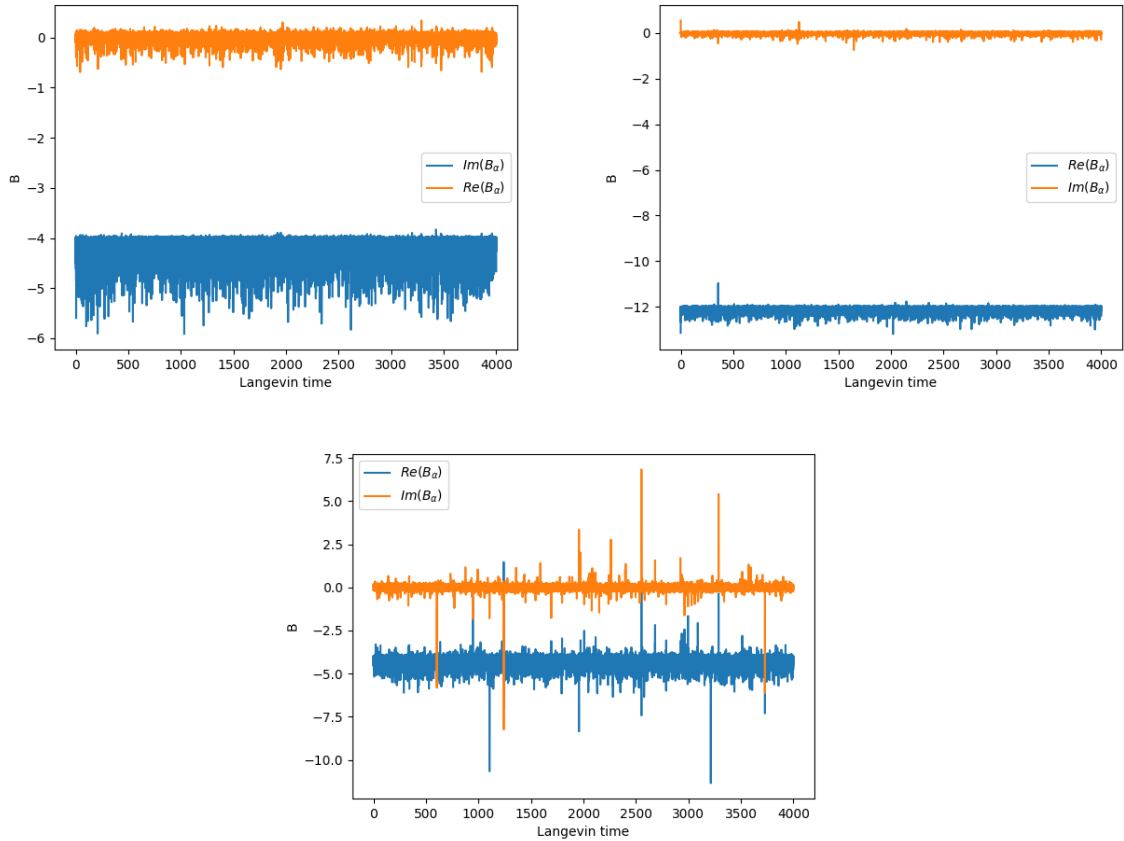


Figure 1: Langevin time history of B . The adaptive Langevin step size was set to be $\Delta t \leq 10^{-3}$ with $N_{\text{gen}} = 10^6$. Measurements were taken after every 50th step.

To check the severity of the sign problem, we have to consider the phase of the Pfaffian, which is the factor multiplying the bosonic action in Eq. (24). So we consider

$$\begin{aligned} P(\alpha, \phi) &= (e^{i\alpha} - 1 + 2g\phi) \\ \implies \frac{P(\alpha, \phi)}{\bar{P}(\alpha, \phi)} &= \frac{(e^{i\alpha} - 1 + 2g\phi)}{(e^{-i\alpha} - 1 + 2g\bar{\phi})} = e^{2i\theta}, \end{aligned} \quad (28)$$

where θ is the phase [8] of the Pfaffian. Our results are summarized in Figs. 2, 3, 4, and 5.

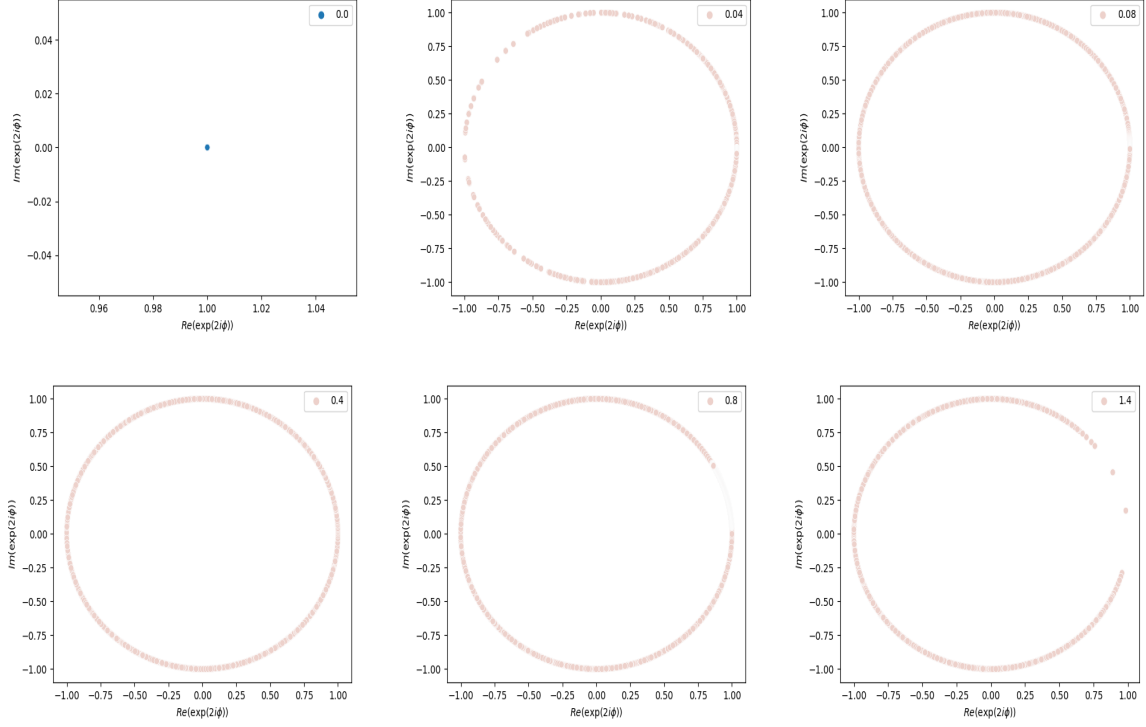


Figure 2: Scatter plot of $\text{Im}(e^{2i\theta})$ vs $\text{Re}(e^{2i\theta})$ for $W(\phi) = g(\mu^2 + \phi^2)$. The adaptive Langevin step size was set to be $\Delta t \leq 4 \times 10^{-3}$ with $N_{\text{gen}} = 10^6$. Measurements were taken after every 50th step.

5.2 $\mathcal{PT}$ Symmetric Models Inspired δ -Potential

Here we consider

$$W(\phi) = -\frac{g}{2+\delta}(i\phi)^{2+\delta} \quad \delta > -2.$$

For this model, we have

$$S'_{\text{eff}} = -ig^2(1+\delta)(i\phi)^{2\delta+1} - \frac{ig\delta(\delta+1)(i\phi)^{\delta-1}}{e^{i\alpha} - 1 + g(1+\delta)(i\phi)^\delta}. \quad (29)$$

The phase factor can be obtained from the Pfaffian

$$\begin{aligned} P(\alpha, \phi) &= (e^{i\alpha} - 1 + g(1+\delta)(i\phi)^\delta) \\ \frac{P(\alpha, \phi)}{\bar{P}(\alpha, \phi)} &= \frac{(e^{i\alpha} - 1 + g(1+\delta)(i\phi)^\delta)}{(e^{-i\alpha} - 1 + g(1+\delta)(-i\bar{\phi})^\delta)} = e^{2i\theta}. \end{aligned} \quad (30)$$

In Fig. 6 we show the Langevin time history of B . In Fig. 7 we show the average of B against Langevin time.

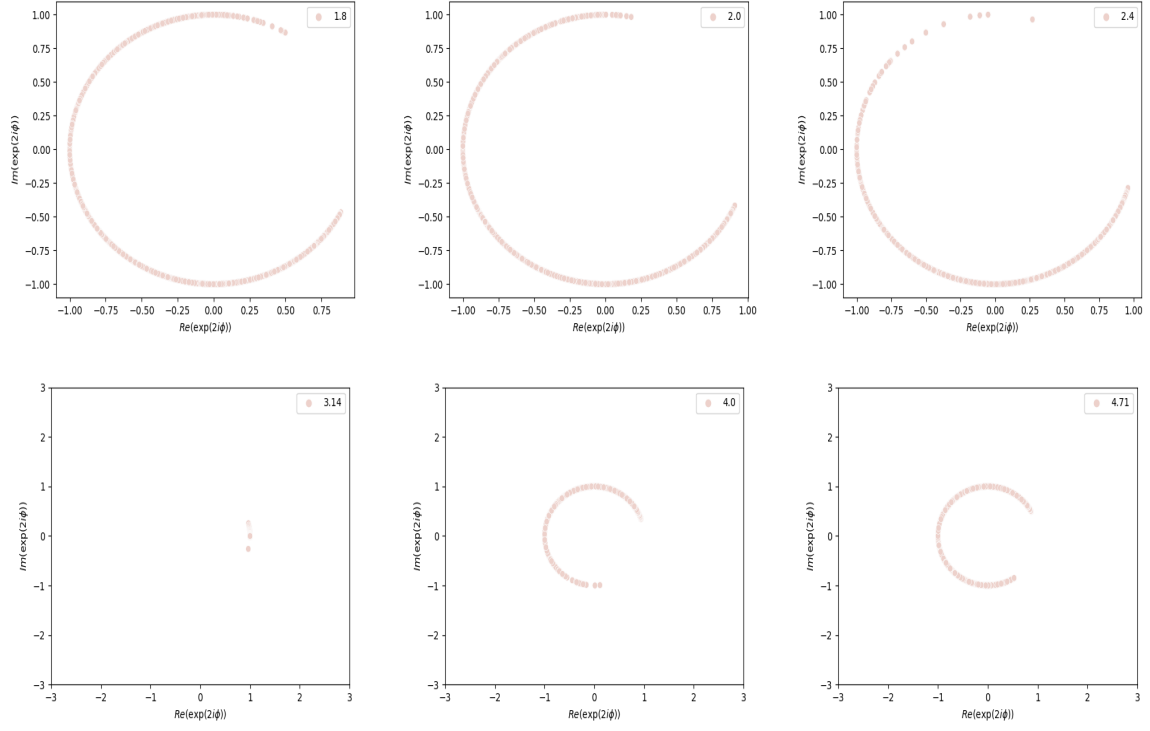


Figure 3: Scatter plot of $\text{Im}(e^{2i\theta})$ vs $\text{Re}(e^{2i\theta})$ for $W'(\phi) = g(\mu^2 + \phi^2)$. The adaptive Langevin step size was set to be $\Delta t \leq 4 \times 10^{-3}$ with $N_{\text{gen}} = 10^6$. Measurements were taken after every 50th step.

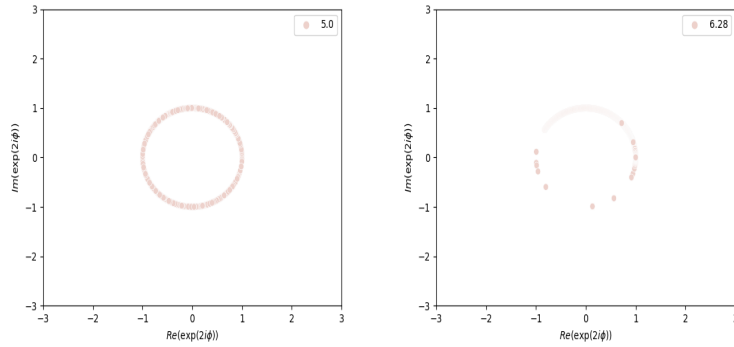


Figure 4: Scatter plot of $\text{Im}(e^{2i\phi})$ vs $\text{Re}(e^{2i\phi})$ for $W'(\phi) = g(\mu^2 + \phi^2)$. The adaptive Langevin step size was set to be $\Delta t \leq 4 \times 10^{-3}$ with $N_{\text{gen}} = 10^6$. Measurements were taken after every 50th step.

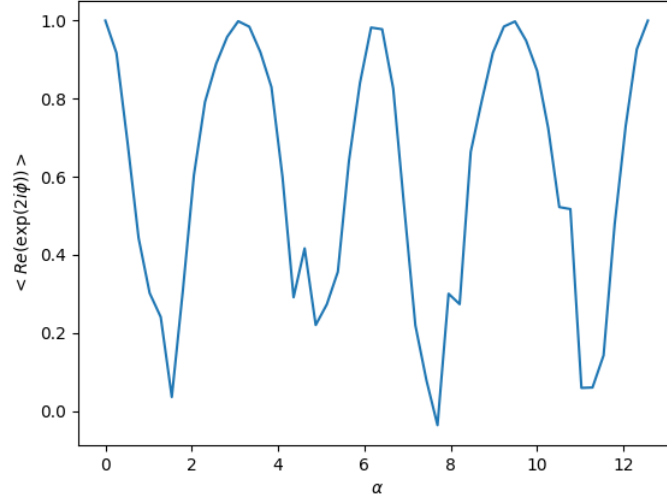


Figure 5: $\text{Re}(e^{2i\theta})$ is plotted against α over the range $[0, 2\pi]$. The adaptive Langevin step size was set to be $\Delta t \leq 4 \times 10^{-3}$ with $N_{\text{gen}} = 10^6$. Measurements were taken after every 50th step.

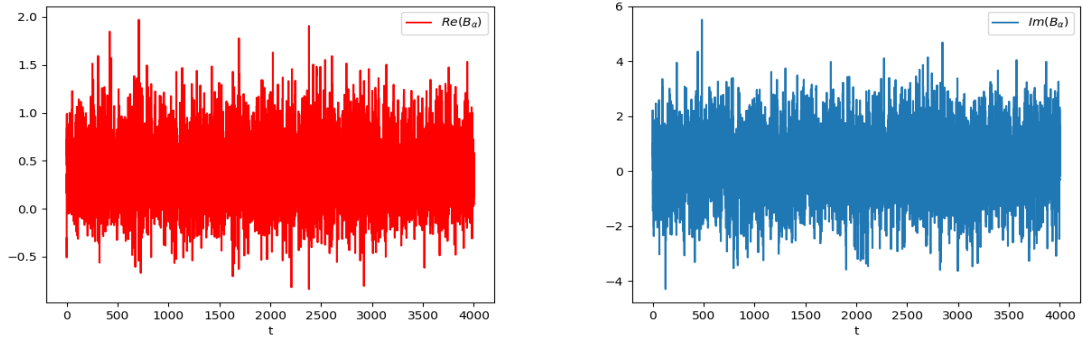


Figure 6: Langevin time history of B . The adaptive Langevin step size was set to be $\Delta t \leq 10^{-3}$ with $N_{\text{gen}} = 10^6$. Measurements were taken after every 50th step.

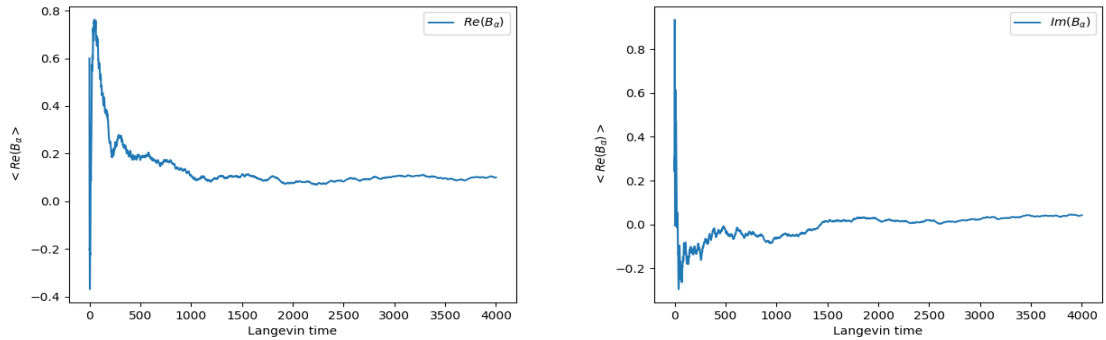


Figure 7: Average of B against Langevin time. The adaptive Langevin step size was set to be $\Delta t \leq 10^{-3}$ with $N_{\text{gen}} = 10^6$. Measurements were taken after every 50th step. The expected value can be calculated by the formulas derived in the previous section.

To check the severity of the sign problem, we consider the phase θ . Our results are summarized in Figs. 8, 9, and 10.

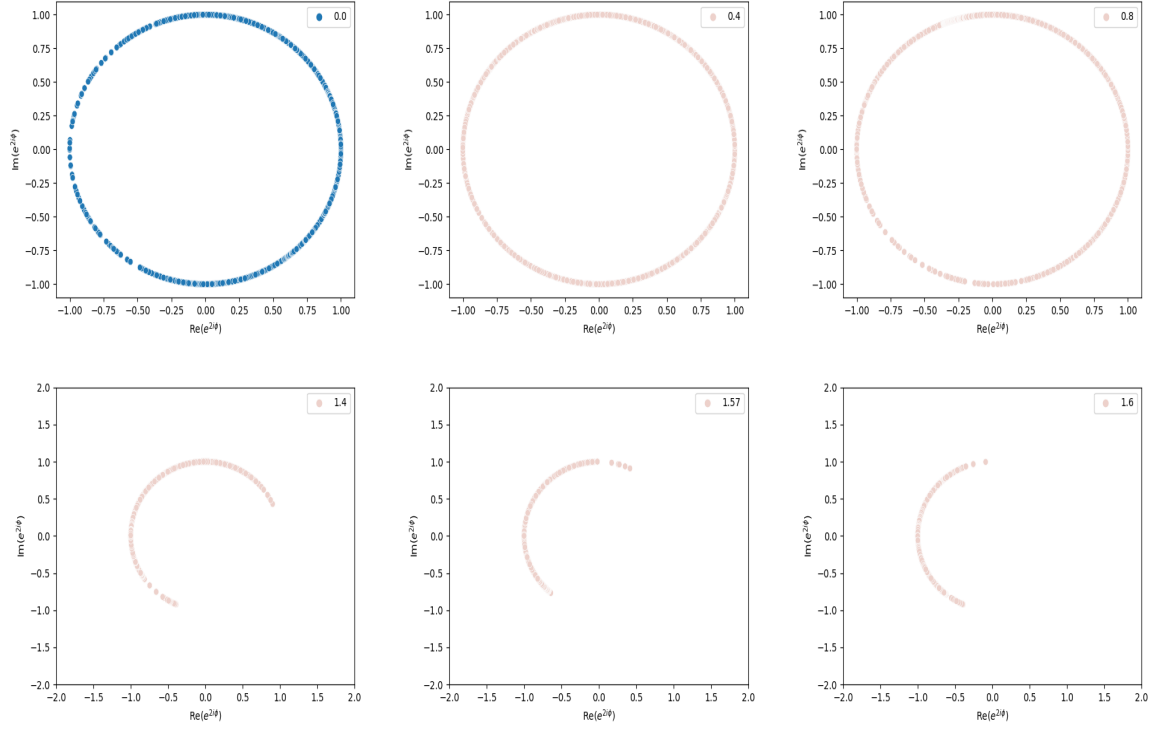


Figure 8: Scatter plot of $\text{Im}(e^{2i\theta})$ vs $\text{Re}(e^{2i\theta})$ for $W(\phi) = -\frac{g}{2+\delta}(i\phi)^{2+\delta}$ for $\delta = 0.5, g = 0.5$. The adaptive Langevin step size was set to be $\Delta t \leq 4 \times 10^{-3}$ with $N_{\text{gen}} = 10^6$. Measurements were taken after every 50th step.

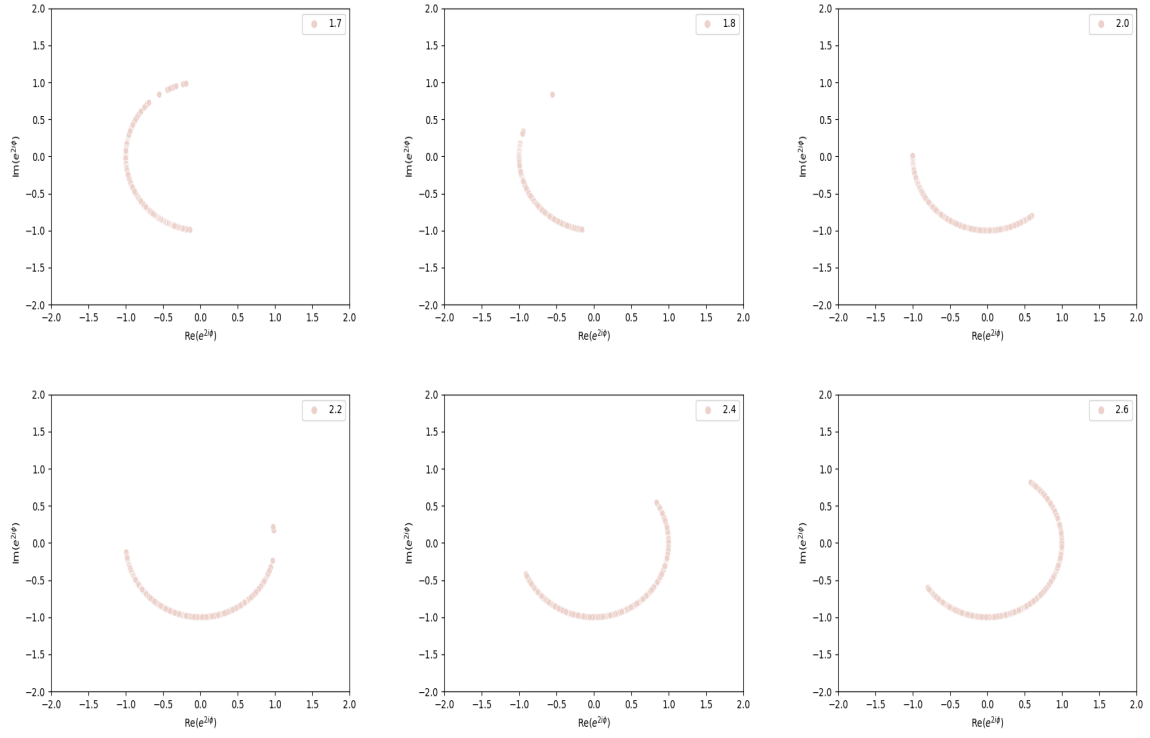


Figure 9: Scatter plot of $\text{Im}(e^{2i\theta})$ vs $\text{Re}(e^{2i\theta})$ for $W(\phi) = -\frac{g}{2+\delta}(i\phi)^{2+\delta}$ for $\delta = 0.5, g = 0.5$. The adaptive Langevin step size was set to be $\Delta t \leq 4 \times 10^{-3}$ with $N_{gen} = 10^6$. Measurements were taken after every 50th step.

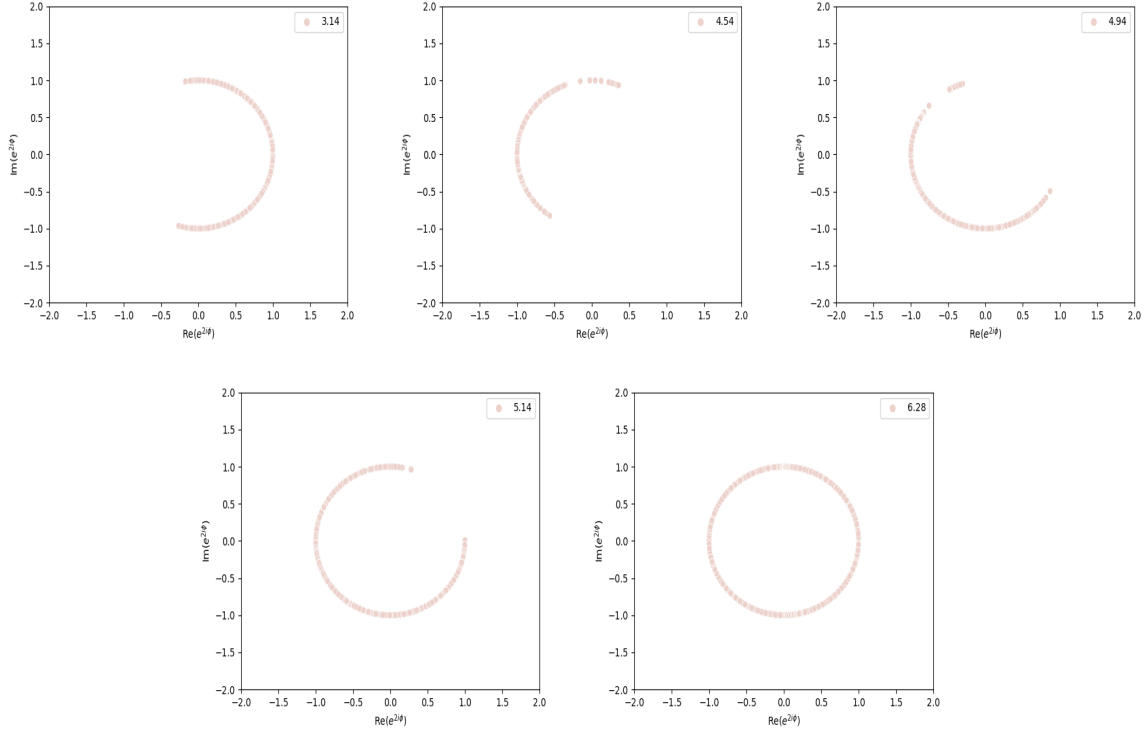


Figure 10: Scatter plot of $\text{Im}(e^{2i\theta})$ vs $\text{Re}(e^{2i\theta})$ for $W(\phi) = -\frac{g}{2+\delta}(i\phi)^{2+\delta}$ for $\delta = 0.5, g = 0.5$. The adaptive Langevin step size was set to be $\Delta t \leq 4 \times 10^{-3}$ with $N_{gen} = 10^6$. Measurements were taken after every 50th step.

Observe that starting from $\alpha = 1.4$, the circular arc of points rotates anti-clockwise with its length decreasing as α increases, and beyond $\alpha > 2$ the length increases again, and at around $\alpha = \pi$ the arc becomes almost a half circle. For $\alpha + \pi$, periodicity in the pattern can be observed. However, this time the length increases with α increasing, and finally, at $\alpha \approx 2\pi$, the points fill the unit circle resembling the pattern for $\alpha = 0$.

We have also checked the dependence of $\text{Re}(\langle \exp(2i\theta) \rangle)$ on α . Our results are summarized in Figs. 11 and 12.

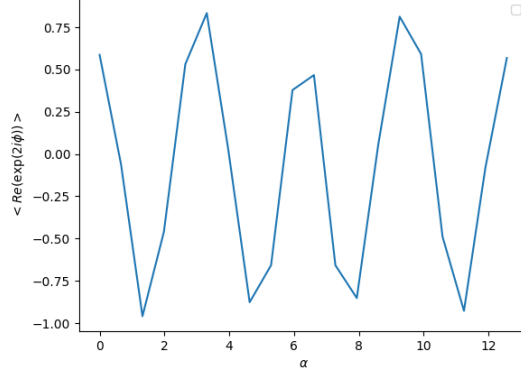


Figure 11: $\text{Re}(\langle e^{2i\theta} \rangle)$ is plotted against α over the range $[0, 4\pi]$ with $g = 0.5$. The adaptive Langevin step size was set to be $\Delta t \leq 4 \times 10^{-3}$ with $N_{gen} = 10^6$. Measurements were taken after every 50th step.

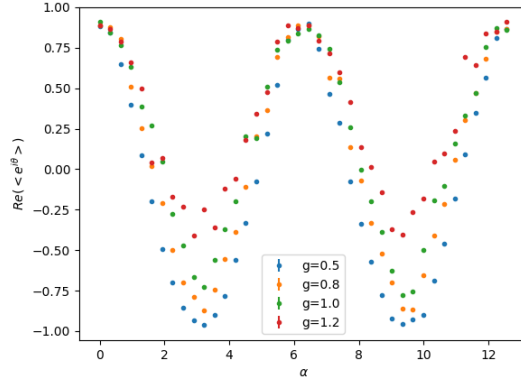


Figure 12: $\text{Re}(\langle e^{i\theta} \rangle)$ is plotted against α over the range $[0, 4\pi]$ for $\delta = 0.5$ with different values of g . The adaptive Langevin step size was set to be $\Delta t \leq 4 \times 10^{-3}$ with $N_{gen} = 10^6$. Measurements were taken after every 50th step.

5.2.1 Reliability of CL Simulations

In this section we attempt to check the reliability of the results obtained from CL simulations using the correctness criterion based on Fokker-Planck equation.

The holomorphic observables of the theory evolve according to [9, 10]

$$\frac{\partial O(\phi, \tau)}{\partial \tau} = \tilde{L}O(\phi, \tau), \quad (31)$$

where

$$\tilde{L} = \left[\frac{\partial}{\partial \phi} - \frac{\partial}{\partial \phi} S(\phi) \right] \frac{\partial}{\partial \phi}.$$

Once the equilibrium distribution is reached its obvious that the time derivative in the above

equation becomes zero and one can have

$$\langle \tilde{L}(O(\phi)) \rangle = 0 \quad (32)$$

The condition above can be used as a *correctness criterion* [9, 10].

We choose the the auxiliary field B as our observable $O(\phi, \tau)$. We have

$$\tilde{L}(B) = -iW''' + iW'W''^2 - \frac{iW''W'''}{(e^{i\alpha} - 1 + W'')}. \quad (33)$$

In Fig.12 observe that the $\text{Re}(\langle e^{i\theta} \rangle)$ becomes 1 in the vicinity of $\alpha = 2n\pi$ for different values of g . So, we check the Langevin time history of $\tilde{L}(B)$ at this values of α for a fixed g in Fig. 13, and also $\langle \tilde{L}(B) \rangle$ in Fig. 14.

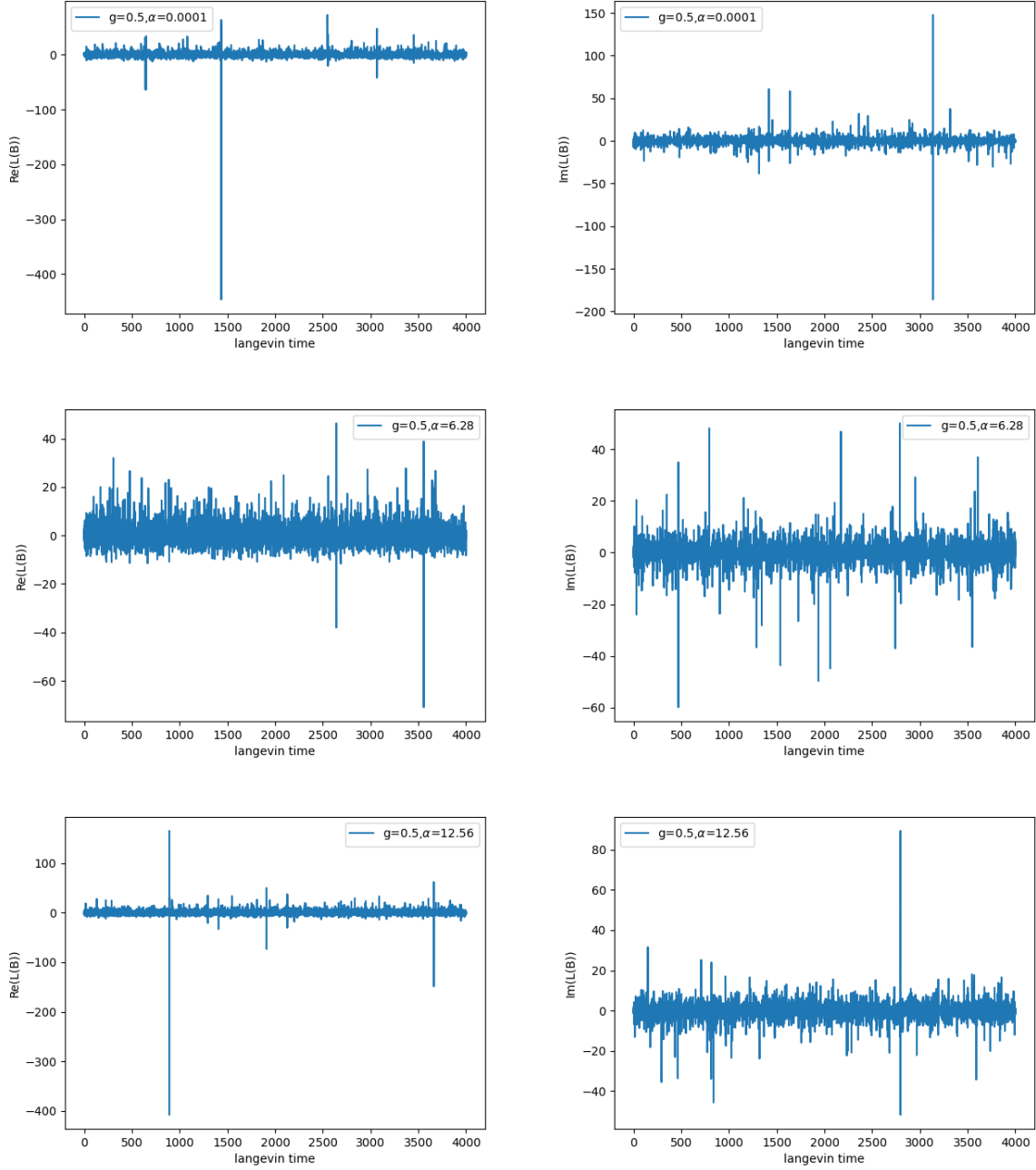


Figure 13: Langevin time history of $\tilde{L}(B)$. The adaptive Langevin step size was set to be $\Delta t \leq 10^{-3}$ with $N_{gen} = 10^6$. Measurements were taken after every 50th step.

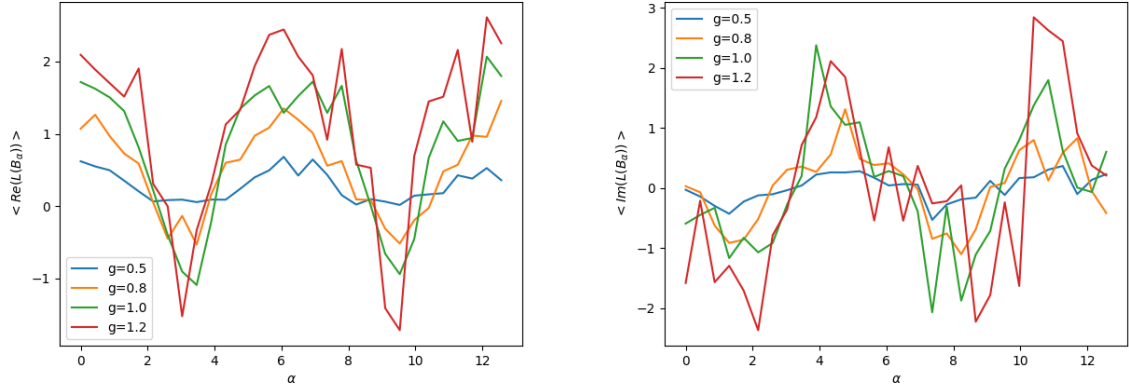


Figure 14: $\text{Re}(\langle \tilde{L}(B) \rangle)$ and $\text{Im}(\langle \tilde{L}(B) \rangle)$ are plotted against α over the range $[0, 4\pi]$ for $\delta = 0.5$ with different values of g . The adaptive Langevin step size was set to be $\Delta t \leq 4 \times 10^{-3}$ with $N_{\text{gen}} = 10^6$. Measurements were taken after every 50th step.

In Fig. 14 we observe that $\text{Re}(\langle \tilde{L}(B) \rangle)$ gets further from zero at $\alpha = 0, 2\pi, 4\pi$ with increasing values of g . Whereas $\text{Im}(\langle \tilde{L}(B) \rangle)$ remains near zero at the same values of α . considering only this criterion we can conclude that the CL simulations are reliable at smaller values of g . There are other correctness criteria [11] that should be considered as well to conclude anything about reliability altogether.

6 Conclusions

We have implemented the complex Langevin method to investigate the severity of the sign problem in $d = 0$ supersymmetric QFTs with real and complex superpotentials. We have also checked the reliability of CL simulations using the correctness-criterion based on the Fokker-Planck equation. We plan to extend our investigations to a range of values of δ in the delta superpotential. In the future, it would be interesting to investigate the severity of the sign problem in $d = 1$ supersymmetric QFTs.

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